SHRS-YOLO: Small Target Detection in Hazy Remote Sensing Images

Runjie Yang, *Student Member, IEEE*, Yongde Guo, Jun Yan, Qiuji Zhou, Haonan Yang, *Student Member, IEEE*, and Dunwei Gong, *Senior Member, IEEE*

***Abstract*—Small target detection in remote sensing (RS) images under hazy conditions faces significant challenges due to atmospheric scattering and absorption phenomena that markedly reduce image contrast and color saturation. Small targets exhibit fragile feature representations, while haze-induced interference complicates target-background discrimination. To address these limitations, we propose small target detection in hazy remote sensing images (SHRSs)-you only look once (YOLO), an end-to-end framework specifically designed for small target detection in hazy RS images. The image dehazing (IDH) module incorporates four coordinated filters: dehazing, low-light enhancement, color compensation, and sharpening. A hyperparameter prediction convolutional neural network (HP-CNN) dynamically determines optimal filter parameters based on input image characteristics. The deep feature fusion (DFF) module uses transmission guidance maps to adaptively fuse multiscale regional attention (MSR-Attention) features with local CNN features, thereby enhancing small target detection capability. Additionally, we introduce an small target detection intersection over union (IoU) loss function with size-based adjustment factors to improve small target detection accuracy. To validate our approach, we synthesized the HazyRS dataset containing 20 000 annotated images with nonuniform haze patterns derived from real satellite imagery. Experimental results demonstrate that SHRS-YOLO achieves 88.37% mAP^{val}_{50} on the HazyRS dataset, representing an 8.40% improvement over baseline methods and a 2.83% improvement over current state-of-the-art (SOTA) approaches.**

***Index Terms*—Attention mechanism, deep feature fusion (DFF), image dehazing (IDH), remote sensing (RS) images, small target detection.**

I. Introduction

With the continuous emergence of high-resolution remote sensing (RS) imagery, deep learning (DL)-based object detection algorithms such as the you only look once (YOLO) series [1], [2], [3] have been widely applied to RS image object detection. However, when RS sensors operate under low visibility conditions (such as haze, dust, or thin clouds), light scattering and absorption phenomena during the imaging process significantly reduce contrast and color saturation, presenting dual challenges to detection models.

1) Small targets occupy an extremely low proportion at the pixel scale, resulting in fragile feature representation.
2) Haze causes texture blurring and noise superposition, making it difficult for models to distinguish targets from backgrounds.

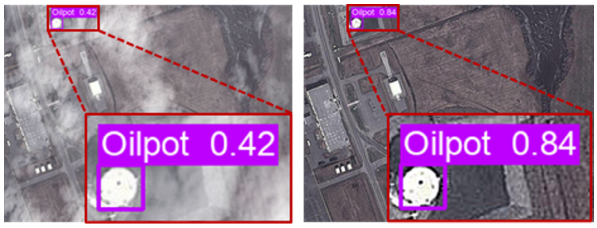


Fig. 1. Under hazy conditions, the proposed method can adaptively enhance image quality, thereby improving the accuracy of detection results.

Fig. 1 demonstrates that directly transferring general detectors like YOLOv11 [3] to hazy RS images reduces detection accuracy. Traditional methods employ sequential pipelines, first processing hazy images through specialized dehazing algorithms, then performing object detection on the restored images. Classical dehazing methods include dark channel prior (DCP) [4], color attenuation prior (CAP) [5], and more recent DL-based approaches [6], [7], [8]. However, these dehazing algorithms are optimized for human visual perception rather than machine-learning tasks; therefore, artifacts and residual noise after dehazing may negatively impact subsequent detection performance. Additionally, independent optimization of dehazing and detection modules leads to inefficient resource utilization.

Some researchers have explored domain adaptation techniques to bridge the gap between clear and hazy image domains. These methods typically treat clear images as the source domain and hazy images as the target domain, employing adversarial training or feature alignment strategies. However, domain adaptation methods face challenges in determining the appropriate adaptation degree for images with varying haze intensities. Furthermore, these methods typically require paired clear-hazy image data, which are difficult to obtain in real RS scenarios.

Recent research has attempted to integrate dehazing and detection functions within unified DL frameworks. Methods

Received 26 August 2025; revised 24 December 2025; accepted 2 January 2026. Date of publication 6 January 2026; date of current version 2 March 2026. This work was supported by the National Natural Science Foundation of China and the Macau Science and Technology Development Fund (NSFC-FDCT) through the Joint Scientific Research Project Fund, Macau, China, under Grant 0051/2022/AFJ. *(Corresponding author: Yongde Guo.)*

Runjie Yang, Yongde Guo, Qiuji Zhou, and Haonan Yang are with the Faculty of Data Science, City University of Macau, Macau, SAR, China (e-mail: D23092100349@cityu.edu.mo; wtkuok@cityu.edu.mo; D22092100031@cityu.edu.mo; D23092100105@cityu.edu.mo).

Jun Yan is with the School of Data Science, Qingdao University of Science and Technology, Qingdao 266000, China, and also with Zhuhai Aerospace Microchips Science and Technology Company Ltd., Zhuhai 519000, China (e-mail: yanjun@qust.edu.cn).

Dunwei Gong is with the School of Data Science, Qingdao University of Science and Technology, Qingdao 266000, China (e-mail: dwgong@vip.163.com).

Digital Object Identifier 10.1109/TGRS.2026.3651676

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

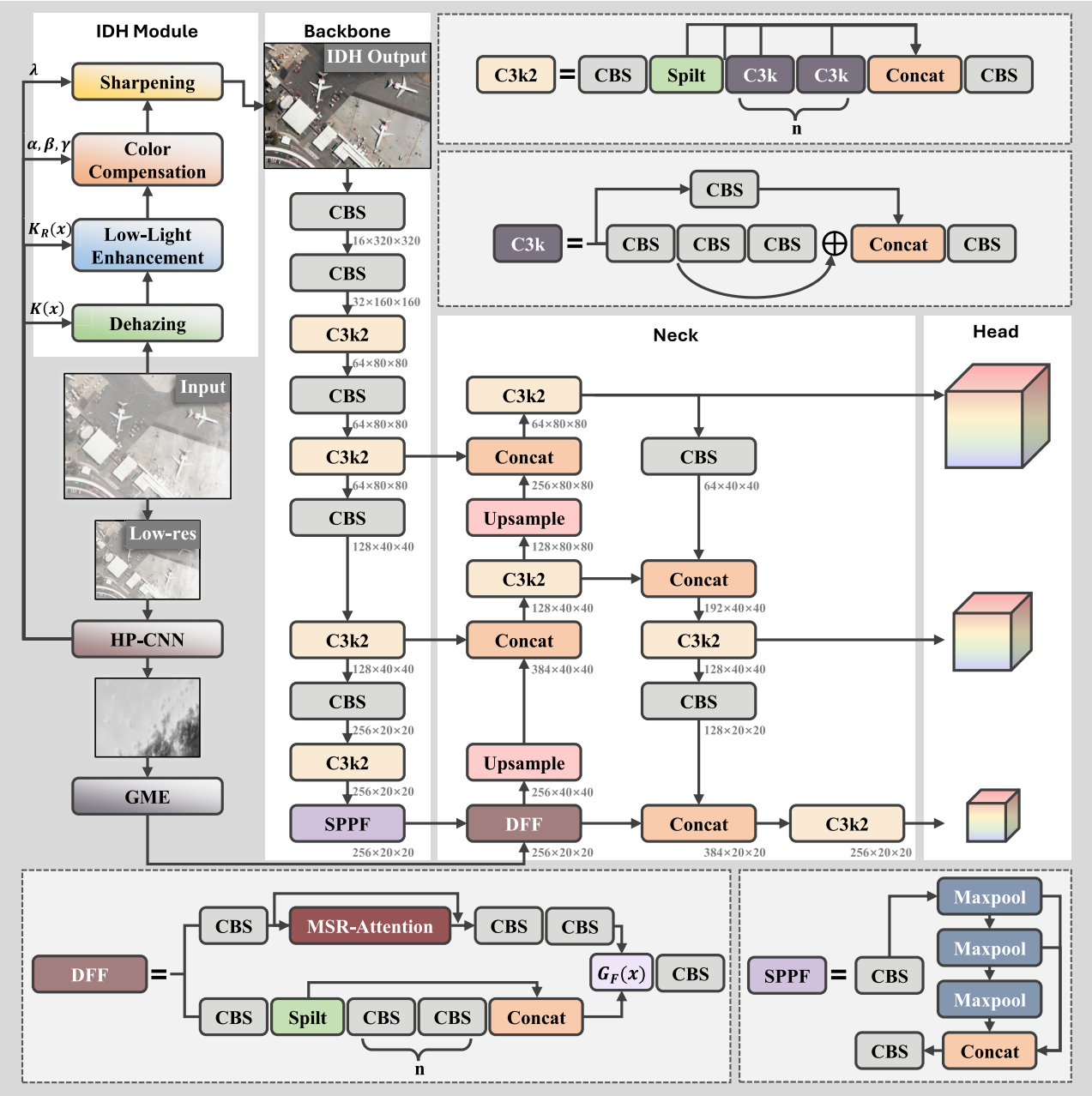


Fig. 2. Comprehensive architecture of the proposed SHRS-YOLO framework, which consists of three key functional components integrated into an end-to-end pipeline. (1) IDH module: This module acts as a preprocessing stage, utilizing a HP-CNN to dynamically regress optimal parameters for four coordinated filters (sharpening, color compensation, low-light enhancement, and dehazing) to restore the input hazy image. (2) GME module: This component extracts a transmission-based guidance map from the IDH module to indicate the spatial distribution of haze density. (3) DFF module: Embedded within the detection backbone, the DFF module leverages the guidance map to adaptively fuse global context captured by MSR-Attention with local features extracted by CNNs. The resulting feature maps are processed through the neck and head to accurately localize small targets.

such as IA-YOLO [9], MS-DAYOLO [10], and DSNet [11] aim to jointly optimize both tasks through shared feature representations. While these methods show promise, they often lack specialized mechanisms for handling the unique characteristics of small targets under nonuniform haze conditions. Moreover, most existing methods do not adequately address the spatial distribution of haze intensity in RS images.

To address these limitations, we propose small target detection in hazy remote sensing images (SHRSs)-YOLO, a novel end-to-end framework specifically designed for small target detection in hazy RS images, as illustrated in Fig. 2. We designed an image dehazing (IDH) module that performs comprehensive image restoration through four coordinated filters: dehazing, low-light enhancement, color compensation, and image sharpening. Unlike fixed-parameter approaches, our system employs a hyperparameter prediction convolutional neural network (HP-CNN) to dynamically determine optimal filter parameters based on input image characteristics. Recognizing the nonuniform nature of haze distribution in real scenarios, we propose a deep feature fusion (DFF) module. The guidance map extraction (GME) component generates transmission-based guidance maps, which are used to adap-

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

tively fuse long-range features efficiently captured by multiscale regional attention (MSR-Attention) with local features captured by CNNs. To enhance small target detection performance, we introduce small target detection intersection over union (STD-IoU), which incorporates size-based adjustment factors to provide greater emphasis on small target detection accuracy. Additionally, we synthesized the hazy remote sensing (HazyRS) dataset, containing 20 000 annotated images with nonuniform haze morphologies from real satellite imagery, providing a benchmark for hazy RS object detection research.

The main contributions of this work are summarized as follows.

1) We proposed SHRS-YOLO, a novel end-to-end framework specifically designed for small target detection in hazy RS images. This framework provides a comprehensive solution for jointly optimizing image enhancement and object detection.
2) We introduced the IDH module for dehazing RS images and enhance the adaptive IDH capability of the IDH module through HP-CNN. Meanwhile, we improve the network's ability to handle haze and small targets through the DFF module embedded in YOLOv11 and the STD-IoU loss function.
3) We constructed the HazyRS dataset, containing 20 000 nonuniform hazy RS images with paired annotations. SHRS-YOLO demonstrates superior performance compared to existing state-of-the-art (SOTA) methods across multiple datasets. Experimental results show that compared to baseline methods, our approach achieves an 8.40% improvement in mAP_{50}^{val} and a 2.83% improvement over current SOTA methods.

The remainder of this article is organized as follows. Section II reviews the related work concerning object detection in RS images and IDH techniques. Section III elaborates on the proposed SHRS-YOLO framework, detailing the IDH module, the HP-CNN, the DFF module, and the STD-IoU loss function. Additionally, the construction process of the HazyRS dataset is described in this section. Section IV presents the experimental setup, ablation studies, and a comprehensive performance comparison against SOTA methods on both synthetic and real-world datasets. Finally, Section V summarizes the conclusions and suggests future research directions.

II. Related Work

A. Target Detection in RS Images

DL-based object detectors automatically extract image features and recognize objects through an end-to-end learning framework. Currently, detection methods can be categorized into two main types: single-stage detectors and two-stage detectors. Compared to two-stage detectors, single-stage detectors offer faster computational speeds and exhibit lower accuracy loss, which enables them to demonstrate greater potential in various application scenarios. The YOLO series, as a classic single-stage object detection algorithm, has significant advantages in the detection of small objects [12], [13]. Recently, several enhanced versions of the YOLO algorithm have emerged in the field of RS object detection, including TPH-YOLOv5 [14], FE-YOLOv5 [15], CA-YOLO [16], and FFCA-YOLO [17].

TPH-YOLOv5 [14] replaces the original prediction head with a transformer prediction head (TPH) that employs a self-attention mechanism to extract richer global contextual information, thereby enhancing its prediction capabilities. Additionally, it integrates a convolutional block attention module (CBAM) to more effectively identify attention regions in scenes with high object density. FE-YOLOv5 [15] utilizes a feature enhancement module (FEM) and a spatial awareness module (SAM). The FEM enhances the discriminative features of small objects through a global attention mechanism and high-level contextual information. The SAM achieves sparse sampling and adaptive spatial learning through deformable convolutions, filtering spatial information and enhancing feature robustness, which allows for improved focus on foreground objects. CA-YOLO [16] introduces a coordinate attention module for shallow feature extraction, establishing long-range dependencies between pixels to effectively suppress redundant backgrounds and enhance the expressive power of target features. FFCA-YOLO [17] is specifically designed for detecting small objects in RS images. This model combines three innovative lightweight modules: the FEM, the feature fusion module (FFM), and the spatial context awareness module (SCAM). The FEM enhances local features of small objects through a multibranch structure, the FFM employs an improved feature fusion strategy to mitigate background interference, and the SCAM utilizes global pooling to learn global contextual relationships between channels.

Beyond the visible spectrum RS domain, recent advancements in object detection under complex scenarios have provided diverse strategies for feature enhancement that are instructive for hazy small target detection. For instance, to address the challenges of poor texture and high noise in infrared (IR) imagery, deep-IRTarget [18] introduced a dual-domain mechanism that effectively allocates resources between spatial and frequency features. To mitigate spectral leakage in IR few-shot scenarios, a frequency compression method [19] was proposed to enhance feature clarity by leveraging localized discrete cosine transforms. In multimodal detection, differential feature awareness networks (DFANet) [20] utilized antagonistic learning to distinctively extract and fuse unique information from IR and visible modalities. Furthermore, for dynamic scenes like sports analysis where targets share similar appearances, deep player identification techniques [21] incorporating pose-guided embeddings have been developed to prevent identity switches during tracking. Collectively, these studies underscore the critical role of specialized feature representation learning when dealing with targets in low-quality or complex backgrounds.

B. Haze Removal in RS Images

Prior-based dehazing methods estimate haze transmission and atmospheric light by exploiting statistical properties of images or introducing external priors to restore scene clarity and true colors [5], [22], [23], [24]. These methods rely on empirical assumptions and therefore often exhibit performance

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

degradation under high-brightness backgrounds, structurally complex scenes, or atypical haze conditions, particularly in RS images. DL has achieved significant progress in RS IDH. Leveraging feature learning capabilities, deep networks can directly map hazy images to clear images in an end-to-end manner, providing more stable and superior dehazing performance than traditional algorithms.

Li et al. [6] proposed a single network called AOD-Net for haze removal, which aims to combine two parameters from an atmospheric scattering model (ASM) into one using a conversion formula. This approach eliminates the intermediary step, resulting in reduced overall error in parameter estimation. The network consists of two modules, one for estimating parameters and the other for generating clear images. These modules work together to achieve the objective of the network. The estimation module determines the network parameters while the generation module produces clear images. Engin et al. [7] proposed a cycle-dehaze system that can generate a dehazed image directly from the input blurry image without requiring any estimation of the ASM parameters. Chen et al. [25] developed a GCANet, which codec utilizes three convolutional blocks as the encoder, while the decoder comprises two convolution blocks and a deconvolution block, separated by hole residual blocks that aggregate contextual data. Many of the existing single-pixel network architectures are incapable of handling ultrahigh definition (UHD) images using a single GPU. Zheng et al. [26] proposed a novel network that combines three deep CNNs, enabling it to dehaze 4K photographs efficiently using a single GPU. This technique utilizes a three-CNN architecture for deblurring: the first network trains a full-resolution bootstrap map to match the learned two-sided model, the second network extracts blur-related features from a reduced-resolution blurred image, and the third network leverages the feature map to generate the deblurred output. Xiao et al. [8] created a model that can immediately interpret 4K hazy images by fusing Taylor's infinite approximation principle with the Laplacian pyramid. The image's high-frequency information is muted by the polynomial.

C. Small Target Detection

Small targets in RS images occupy a minimal proportion of the entire image and are susceptible to interference from complex geographical backgrounds and atmospheric conditions. With the rapid development of DL technology, new-generation YOLO detectors targeting aerial RS scenarios and tiny target recognition challenges have continuously emerged.

Among these innovative solutions, ESOD-YOLO [27] designed a cross-hierarchical multiscale fusion architecture with waveform feature pyramids, aiming to enhance the synergy between spatial and semantic information. This model is specifically optimized for small-sized targets, integrating dedicated detection heads and employing a triple focal loss strategy to effectively address the common class distribution imbalance problem in aerial data. Regarding feature processing, RTSOD-YOLO [28] focuses on enhancing the adaptive performance of the Adown component and is equipped with adaptive spatial attention mechanisms to optimize feature representation after downsampling, achieving precise localization of important regions. This method also adopts a ternary feature encoding strategy that combines shallow-layer scale sequence information with dual-encoded features through cascaded serial and parallel feature pathways, providing richer fine-grained information for small-scale target detection. RE-YOLO [29] embeds spatial extraction attention components at key bottleneck positions in the backbone structure, enabling the learning of more discriminative feature representations and strengthening semantic understanding. Meanwhile, it adopts a three-path aggregation network architecture as the neck structure to promote deep fusion of shallow spatial information and deep abstract information, thereby enhancing the model's scene understanding capability. For the core issue of multiscale feature integration, Gold-YOLO [30] proposes an innovative aggregation–distribution strategy that fully integrates convolutional operations with self-attention mechanisms, achieving effective unification of cross-scale features. Regarding lightweight design, EAL-YOLO [31] exemplifies the development direction of attention-driven lightweight detectors. This model reconstructs the backbone network based on EfficientFormerV2, integrating large kernel separable convolution attention with spatial pyramid pooling technology. Through the ASF2-Neck structure, it enhances the perception capability for tiny targets and adopts a lightweight shared convolutional detection head design, significantly reducing computational overhead while maintaining high accuracy.

III. Proposed Method

A. Overview

Under hazy weather conditions, RS images captured by satellites and airborne equipment are susceptible to interference from specific environmental factors, which ultimately results in reduced visibility and decreased target detection accuracy. To address this challenge, we propose a small object detection framework for RS images designed to improve detection accuracy by mitigating the effects of hazy weather. Fig. 2 presents a comprehensive overview of this framework, which incorporates the HP-CNN, IDH module, and SHRS-YOLO component.

Initially, the input image is resized to 256 $\times$ 256 pixels and fed into the HP-CNN to predict the filter parameters for the IDH module. The resulting dehazed image is then used as the input for the SHRS-YOLO detector. Distinct from existing joint optimization methods [9], [10], which typically process dehazed images using standard detection backbones without explicitly leveraging haze density information, our approach introduces a novel interaction mechanism. Specifically, the DFF module embedded in the backbone utilizes the transmission map generated by the IDH module as a guidance signal. This allows the network to adaptively weigh the contributions of local features captured, which are effective for regions with light haze, and global features extracted, which are essential for recovering details in heavily hazy areas. This mechanism is specifically designed to address nonuniform haze distribution in RS images, enabling the detector to effectively integrate long-range and local features for the accurate and efficient detection of small targets.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

B. IDH Module

Hazy RS images suffer not only from information loss but also from reduced brightness, color distortion, and blurred details. Therefore, it is necessary to perform dehazing on these images along with low-light enhancement, color correction, and sharpening operations. The proposed IDH module comprises four adjustable parameter filters: the IDH filter, low-light enhancement filter, color compensation filter, and image sharpening filter.

1) IDH Filter: The impact of haze on RS imaging primarily occurs through light scattering and absorption, significantly degrading image quality. Hazy RS images are formed based on the ASM [32], which allows for the recovery of clear images by estimating the atmospheric light and the medium transmission rate. According to this model, a hazy image can be represented by the following formula:

$$I(x) = J(x)\,t(x) + A\,(1 - t(x)) \tag{1}$$

where $I(x)$ represents the hazy image and $J(x)$ denotes the clear image representation. A is the global atmospheric light value, while $t(x)$ is the medium transmission rate map, defined as follows:

$$t(x) = e^{-\beta} d(x) \tag{2}$$

where β is the atmospheric scattering coefficient and $d(x)$ represents the scene depth.

Therefore, for a given hazy image $I(x)$, it is sufficient to estimate the global atmospheric light value A and the medium transmission rate map $t(x)$. Once A and $t(x)$ are obtained, they can be substituted into the ASM to compute the dehazed image $J(x)$. To minimize the number of parameters that need to be estimated, the following transformation of the ASM is based on existing literature [6]:

$$J(x) = K(x)\,I(x) - K(x) + b \tag{3}$$

and

$$K(x) = \frac{\frac{1}{t(x)}\,(I(x) - A) + (A - b)}{I(x) - 1}. \tag{4}$$

In this way, $t(x)$ and A are reduced to a new variable $K(x)$. The constant bias b is set to a default value of 1. Consequently, we only need to estimate the value of $K(x)$ as a hyperparameter of the dehazing filter, restoring the hazy image $I(x)$ to a haze-free image $J(x)$.

2) Low-Light Enhancement Filter: Camera systems on satellites and drones are typically employed under various lighting and weather conditions. However, they may not be optimized for low-light environments, which often results in dark images, adversely affecting detection accuracy. Therefore, hazy RS images also require low-light enhancement processing. Research [33], [34] indicates that inverse low-light images share similarities with hazy images, suggesting that an inverse operation should be performed as the initial step of low-light enhancement

$$J_R(x) = 255 - J(x). \tag{5}$$

Here, $J(x)$ is the dehazed image output from the dehazing filter, while $J_R(x)$ denotes the image obtained by pixel inversion.

Following the processing steps of the IDH filter, we apply dehazing to the inverted image:

$$L_R(x) = K_R(x)\,J_R(x) - K_R(x) + b_R \tag{6}$$

and

$$K_R(x) = \frac{\frac{1}{t_R(x)}\,(J_R(x) - A_R) + (A_R - b_R)}{J_R(x) - 1} \tag{7}$$

where $L_R(x)$ is the image obtained after applying the inversion-based dehazing technique, $t_R(x)$ denotes the transmission map of the inverted image, and A_R is the global atmospheric light value associated with the inverted image. Additionally, b_R is a constant term set to a default value of 1.

The estimation method for $K_R(x)$ is the same as that of the dehazing filter. By substituting $K_R(x)$ into (6), we can obtain the dehazed output of the inverted image $L_R(x)$. Finally, by performing another inversion operation on the dehazed result $L_R(x)$, we can achieve an image with enhanced visibility under low-light conditions

$$R(x) = 255 - L_R(x). \tag{8}$$

$R(x)$ represents the image after low-light enhancement, where the hyperparameter $K_R(x)$ obtained through HP-CNN can be used for low-light image filtering enhancement.

3) Color Compensation Filter: Specific spectral interference is a common phenomenon during the processes of IDH and low-light enhancement. Additionally, hazy RS images captured under different weather or climate conditions may exhibit color distortions. To mitigate the color distortion that occurs when generating haze-free images, employing a color compensation filter is essential [35]. This filter plays a crucial role in compensating for color loss in dehazed images and reconstructing color information [36]. The color compensation process can be expressed as follows:

$$C(x) = \alpha R_r(x) + \beta R_g(x) + \gamma R_b(x) \tag{9}$$

where $R_r(x)$, $R_g(x)$, and $R_b(x)$ represent the pixel values of the RGB channels, while α, β, and γ are the scaling factors for the RGB channels.

By estimating the values of α, β, and γ with HP-CNN, we can fine-tune the pixel values of the RGB channels to achieve color correction in the image.

4) Image Sharpening Filter: Image sharpening techniques effectively enhance the clarity of object edges and improve the overall definition of an image. Given the correlation of high-frequency components with noise, enhancing details may unintentionally amplify noise. After applying sharpening techniques, noise pixels in the image may impact the accuracy of object detection [37]. Therefore, Gaussian blur needs to be applied first, followed by sharpening operations on the RS image:

$$S(x) = C(x) + \lambda\,(C(x) - \text{Gauss}(C(x))) \tag{10}$$

where $C(x)$ is the input image, Gauss() is the Gaussian filter, and λ is an adjustment factor. By optimizing the value of λ, the degree of sharpening can be improved to enhance detection performance. Our HP-CNN network is capable of generating suitable λ values based on the input image.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

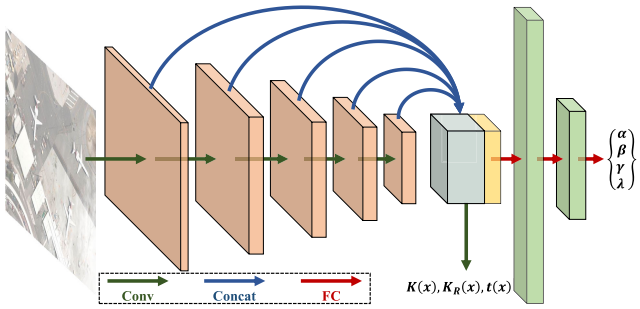


Fig. 3. HP-CNN network consists of five convolutional blocks, a concatenation layer, and two fully connected layers.

C. HP-CNN

In image enhancement, employing adjustable filters necessitates that experienced engineers manually tune hyperparameters [38]. This method has proven both costly and inefficient in complex and variable scenarios. To address this challenge, we designed a compact HP-CNN as a parameter prediction tool. HP-CNN automatically determines suitable filter hyperparameters by analyzing the diverse scene information encoded in the image, thereby effectively enhancing the quality of foggy RS images.

Using hazy scenes as an example, HP-CNN predicts the parameters of the dehazing filter by examining the overall image information, including brightness, color, and hue, as well as the severity of haze. Consequently, even downsampled small images can effectively capture this information, significantly reducing computational costs. For input images of any resolution, we employ bilinear interpolation to scale them to a resolution of 256×256. As shown in Fig. 3, the HP-CNN network comprises five convolutional modules, one concatenation layer, and two fully connected layers. Each convolutional module consists of a 3×3 convolution layer with a stride of 2, followed by a leaky ReLU layer. The final fully connected output layer provides the hyperparameters for the color compensation filter and the image sharpening filter. The output channels of the five convolutional layers are 16, 32, 32, 32, and 32, respectively.

D. DFF Module

To fully leverage the potential of joint optimization, the DFF module is designed to maximize data richness during feature extraction. In traditional sequential approaches, the dehazing step typically collapses the complex atmospheric scattering process into a single restored image, inevitably discarding valuable environmental context such as the spatial distribution of haze density. In contrast, our DFF module explicitly integrates physical priors into the detection loop. Specifically, the transmission map generated by the IDH module is utilized not merely as an intermediate product but as a critical guidance signal. This enables the detector to access a richer information space—combining visible texture features with atmospheric depth cues—thereby facilitating more robust representation learning for small targets that are otherwise easily overwhelmed by noise. To achieve this effective fusion and extraction, the DFF module consists of the following key components.

1) MSR-Attention: Objects in RS images often occupy small spatial regions and may be obscured by complex background structures. Previous approaches primarily processed input features by dividing them into fixed regions, limiting the model's ability to capture information at different scales and potentially hindering the simultaneous recognition of small and large targets. To effectively extract feature information at various scales, we propose an MSR-Attention mechanism. MSR-Attention not only extracts feature information at multiple scales but also dynamically selects relevant regions when processing different queries, which is particularly crucial for detecting small targets.

Given an input feature map $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, the feature map is processed at multiple scales $S_1, S_2, \ldots, S_n$, which can be obtained through downsampling or convolutions of varying sizes. For each scale S_i, the input feature map $\mathbf{X}_i$ is divided into nonoverlapping regions, each containing $(HW)/(S_i^2)$ feature vectors. For each scale S_i, regional queries, keys, and values are computed as follows:

$$\mathbf{Q}_i^r = \mathbf{X}_i^r \mathbf{W}_{q_i}, \quad \mathbf{K}_i^r = \mathbf{X}_i^r \mathbf{W}_{k_i}, \quad \mathbf{V}_i^r = \mathbf{X}_i^r \mathbf{W}_{v_i} \tag{11}$$

where $\mathbf{Q}_i^r$ represents the region-level query feature matrix, $\mathbf{K}_i^r$ denotes the region-level key feature matrix, and $\mathbf{W}_{q_i}$, $\mathbf{W}_{k_i}$, and $\mathbf{W}_{v_i}$ are the weight matrices for queries, keys, and values, respectively. Each query token $\mathbf{Q}_i^r$ is scored using a multilayer perceptron (MLP) to assess its importance relative to each region

$$s_{i,j}^r = \mathrm{MLP}\left(\mathbf{Q}_{i,j}^r\right) \tag{12}$$

where MLP() refers to a fully connected network. This process enables the model to capture feature associations between the jth query token and various regions at scale S_i. Based on the calculated scores, the softmax function generates the regional weight matrix w_i^r at scale S_i, indicating the relative importance of different regions to the query token

$$w_i^r\left[j\right] = \mathrm{Softmax}\left(s_{i,j}^r\right) \tag{13}$$

where $w_i^r[j]$ denotes the contribution of the jth region to the query token. Using the weight matrix w_i^r, we dynamically select the corresponding keys and values from the regions

$$\mathbf{K}_i^w = \sum_{j=1}^{S^2} w_i^r\left[j\right] \mathbf{K}_i^r\left[j\right] \tag{14}$$

$$\mathbf{V}_i^w = \sum_{j=1}^{S^2} w_i^r\left[j\right] \mathbf{V}_i^r\left[j\right] \tag{15}$$

where $\mathbf{K}_i^r[j]$ and $\mathbf{V}_i^r[j]$ represent the key and value features of the jth region, respectively. Subsequently, attention calculations are performed between tokens within the selected regions at scale S_i

$$O_i = \mathrm{Attention}\left(\mathbf{Q}_i^r, \mathbf{K}_i^w, \mathbf{V}_i^w\right) + \mathrm{LCE}\left(\mathbf{V}_i^r\right) \tag{16}$$

where $\mathrm{LCE}(\mathbf{V}_i^r)$ is a local context enhancement term implemented using depthwise separable convolutions. The attention calculation is given by

$$\mathrm{Attention}\left(\mathbf{Q}_i^r, \mathbf{K}_i^w, \mathbf{V}_i^w\right) = \mathrm{Softmax}\left(\frac{\mathbf{Q}_i^r \mathbf{K}_i^{wT}}{\sqrt{d_k}}\right) \mathbf{V}_i^w \tag{17}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

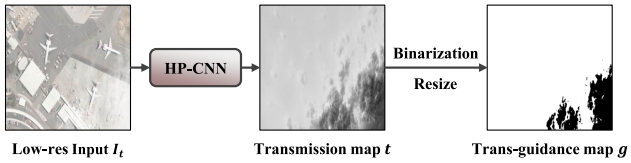


Fig. 4. Trans-GME module.

where d_k is the dimension of the keys, ensuring stability during training and preventing excessively large values. The multi-scale strategy of MSR-Attention enhances the model's ability to capture information at different spatial scales, improving recognition capabilities for targets of various sizes. Furthermore, dynamically selecting relevant regions at each scale enables the model to focus more effectively on potential small target areas.

2) GME Module: The transmission map of an image indicates haze levels and represents the transmission of light through the haze at each pixel. This transmission is related to the concentration and density of the medium through which the light passes. To mitigate the impact of noise in hazy areas on information reconstruction and to enhance overall image quality, we utilized a guided map containing transmission information for feature fusion, as shown in Fig. 4. We designed a transmission-guided map extraction module to capture feature information related to haze intensity in hazy images.

First, the transmission map is denoted as $t \in \mathbb{R}^{H\times W}$ and is initially extracted from the resized small image $I_t \in \mathbb{R}^{H\times W\times 3}$ using HP-CNN. Next, an appropriate threshold is selected to binarize the adjusted $t \in \mathbb{R}^{H\times W}$, resulting in the transmission-guided map $g \in \mathbb{R}^{H\times W}$

$$g(x) = \begin{cases} 1, & t(x) > a \\ 0, & t(x) \le a \end{cases} \tag{18}$$

where x refers to the position of a pixel on the transmission map. The transmission direction map $g \in \mathbb{R}^{H\times W}$ reflects the level of haze in the input image.

3) Transmission-Guided Feature Fusion: The proposed SHRS-YOLO network is developed by integrating a multiscale attention mechanism into the YOLOv11 backbone. Specifically, the DFF module is incorporated within the backbone to replace the C2PSA module, facilitating the fusion of deep features. For hazy RS images, the complete extraction and effective fusion of local and global features are critical for enhancing target detection accuracy. The ASM theory posits that the reasons an image sensor captures data can be attributed primarily to two factors.

1) Particles suspended in the atmosphere cause energy attenuation, reducing the contrast and brightness of the imaging system. These particles can absorb and scatter the target light.
2) The atmospheric medium scatters ambient light, resulting in increased background illumination.

Consequently, hazy images are prone to information loss and noise. In actual hazy images, the concentration distribution of haze is often uneven; compared to areas with light haze, regions with dense haze experience more significant information loss and elevated noise levels. In lightly hazy areas, local information is often sufficient for effective image recovery. Conversely, in areas with dense haze, substantial information loss necessitates the use of global features to aid recovery. However, relying exclusively on global information to reconstruct hazy areas may result in noise in dense haze interfering with the recovery of regions with light haze. Addressing these challenges, the DFF module is designed with two branches to extract both local and global information, employing a feature-guided map to direct the information fusion. As depicted in Fig. 2, the initially extracted features are input into the DFF module. On one hand, the initial features are fed into the global feature extraction branch located in the MSR-Attention module. Through regional feature selection, global information $F_G \in \mathbb{R}^{h\times w\times C}$ is extracted across different spatial scales. Concurrently, the initial features also enter a Pconv layer to fully extract local features $F_L \in \mathbb{R}^{h\times w\times C}$. Fig. 5 illustrates the combination of local features $F_L \in \mathbb{R}^{h\times w\times C}$ (in red) and global features $F_G \in \mathbb{R}^{h\times w\times C}$ (in green) obtained by the feature extraction module. The extracted global features and local features are then fully fused using the transmission-guided map $F \in \mathbb{R}^{h\times w\times C}$

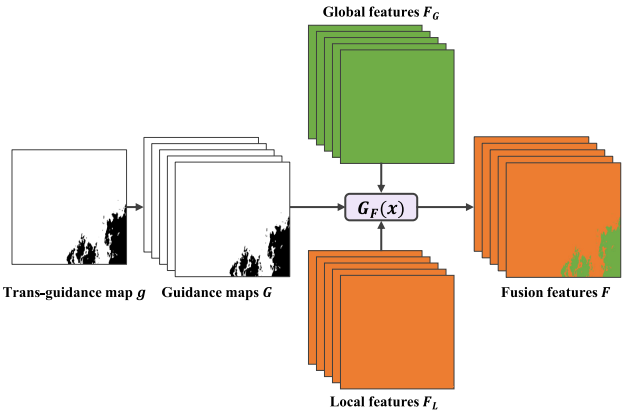


Fig. 5. DFF module.

$$F = F_L \odot \left(\mathrm{Conv}_{3\times 3}(g)\right) + F_G \odot \left(1 - \left(\mathrm{Conv}_{3\times 3}(g)\right)\right) \tag{19}$$

where $\mathrm{Conv}_{3\times 3}()$ refers to the 3×3 convolution, and $\odot$ denotes element-wise multiplication.

E. STD-IoU

The currently most widely used bounding box regression loss function is the CIoU loss [2]. CIoU loss considers three critical geometric factors: the overlapping area, the distance between center points, and the aspect ratio

$$\mathcal{L}_{\mathrm{CIoU}} = 1 - \mathrm{IoU} + \frac{\rho^2}{c^2} + \alpha v \tag{20}$$

where IoU indicates the intersection over union (IoU) of the predicted bounding box and the ground truth bounding box. The term ρ denotes the Euclidean distance between the center points of the predicted box and the true box, while c represents the length of the diagonal of the minimum enclosing rectangle formed by the predicted box and the true box. The parameter

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

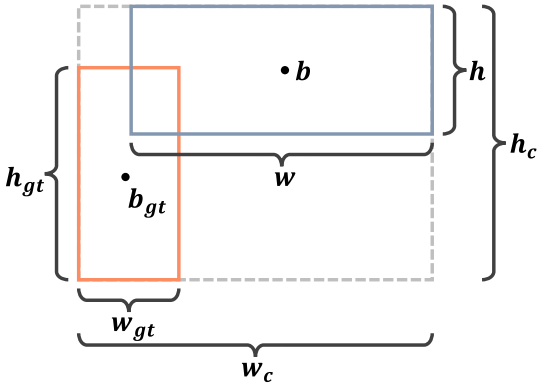


Fig. 6. STD-IoU calculation diagram.

α is a weighting factor and v represents the similarity of the aspect ratio, which can be expressed as follows:

$$v = \frac{4}{\pi^2}\left(\arctan\frac{w_{gt}}{h_{gt}} - \arctan\frac{w}{h}\right)^2 \tag{21}$$

where w_{gt} and h_{gt} represent the width and height of the ground truth box, respectively, while w and h denote the width and height of the predicted box, as illustrated in Fig. 6. To enhance the performance of the loss function when detecting small targets, we propose a new loss function termed STD-IoU, which introduces a size-based adjustment factor S. This factor is calculated by examining the area difference between the predicted box and the ground truth box. Given that small targets typically occupy fewer pixels in the image, CIoU loss may lead to significant errors in detecting these small targets. By incorporating S, STD-IoU places greater emphasis on loss adjustments when the area of small targets changes, thereby enabling the model to better accommodate the detection of small targets

$$\mathcal{L}_{\text{STD-IoU}} = 1 - \text{IoU} + \frac{\rho^2}{c^2} + \alpha v - \beta S \tag{22}$$

and

$$S = \left|\frac{w_{gt}h_{gt} - wh}{w_{gt}h_{gt} + \epsilon}\right| \tag{23}$$

where ϵ is a small constant to avoid division by zero. STD-IoU introduces the hyperparameter β into the loss function, facilitating adjustments to the impact of the loss based on target size. By modifying β, researchers can enhance the recognition capabilities for small targets according to specific application needs. This flexibility allows STD-IoU to be optimized for various datasets or application scenarios.

F. HazyRS Dataset

Effectively training the parameters of deep neural networks typically requires substantial amounts of data. However, in real-world scenarios, obtaining hazy RS images captured by satellites and aerial systems, along with their corresponding clear images, is particularly challenging. To validate the effectiveness of our algorithm, we synthesized a nonuniform hazy RS image object detection dataset based on two existing datasets: the LEarning, VIsion, and remote (LEVIR) dataset [39] and the vehicle detection in aerial imagery (VEDAI) dataset [40].

1) The LEVIR dataset contains over 22 000 high-resolution images sourced from Google Earth, each having a resolution of 800 × 600 pixels and a pixel density ranging from 0.2 to 1.0 m. This dataset encompasses various ground features in human environments, including urban, rural, mountainous, and marine areas, while excluding extreme terrains such as glaciers, deserts, and gobi regions. The LEVIR dataset primarily focuses on three target types: airplane, ship, and oilpot. A total of 11 000 independent bounding boxes (BB) have been annotated within the dataset, with 4724 labeled as airplanes, 3025 as ships, and 3279 as oilpots, yielding an average of 0.5 targets per image.
2) The VEDAI dataset is specifically designed for small object detection tasks, consisting of cropped images obtained from a larger dataset provided by the Utah Automated Geographic Reference Center (AGRC). This dataset features both RGB and IR images for the same scenes; however, we selected only the RGB images for our experiments. It includes over 1200 high-resolution aerial images, divided into two resolutions (1024 × 1024 and 512 × 512). The dataset categorizes vehicles into nine distinct types, including airplanes, boats, RVs, sedans, pickups, tractors, trucks, vans, and others. Each image contains an average of 5.5 vehicles, accounting for approximately 0.7% of the total pixel count within the images. In total, over 3700 targets are annotated in the dataset, which features diverse backgrounds, covering urban environments, roads, and construction sites.

We randomly selected 20 000 images from the LEVIR and VEDAI datasets as clear images, as illustrated in Fig. 7(a). To accurately recreate real-world nonuniform haze patterns, we obtained four satellite RS images with varying haze coverage from the Landsat8 OLI-TIRS [41] and cropped them into 1000 nonoverlapping images. From these cropped images, we randomly selected 200 nonuniform hazy images, as depicted in Fig. 7(b). The transmittance map for each selected image was computed using DCP [4], as shown in Fig. 7(c). To simulate hazy images at different concentrations, we enhanced the extracted transmittance maps across 100 levels, following the equation:

$$t_{i+1}(x) = \begin{cases} t_i(x) + a, & t_i(x) \le 1 - a \\ 1, & t_i(x) > 1 - a \end{cases} \tag{24}$$

where $i \in [0, 99]$ and t_0 is the originally selected transmittance map. The adjustment factor a is set at $a = 0.01$. After adjusting the transmittance levels, we obtained 20 000 transmittance maps with varying shapes and concentrations of haze. Utilizing the ASM, we combined the clear images with the transmittance maps to create nonuniform hazy RS images, as illustrated in Fig. 7(d). Our synthesized HazyRS dataset comprises a total of 20 000 hazy images, their corresponding clear images, transmittance maps, and annotations. The types of BB and their specific details within the dataset are illustrated in Fig. 8. To validate the small object detection capability of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

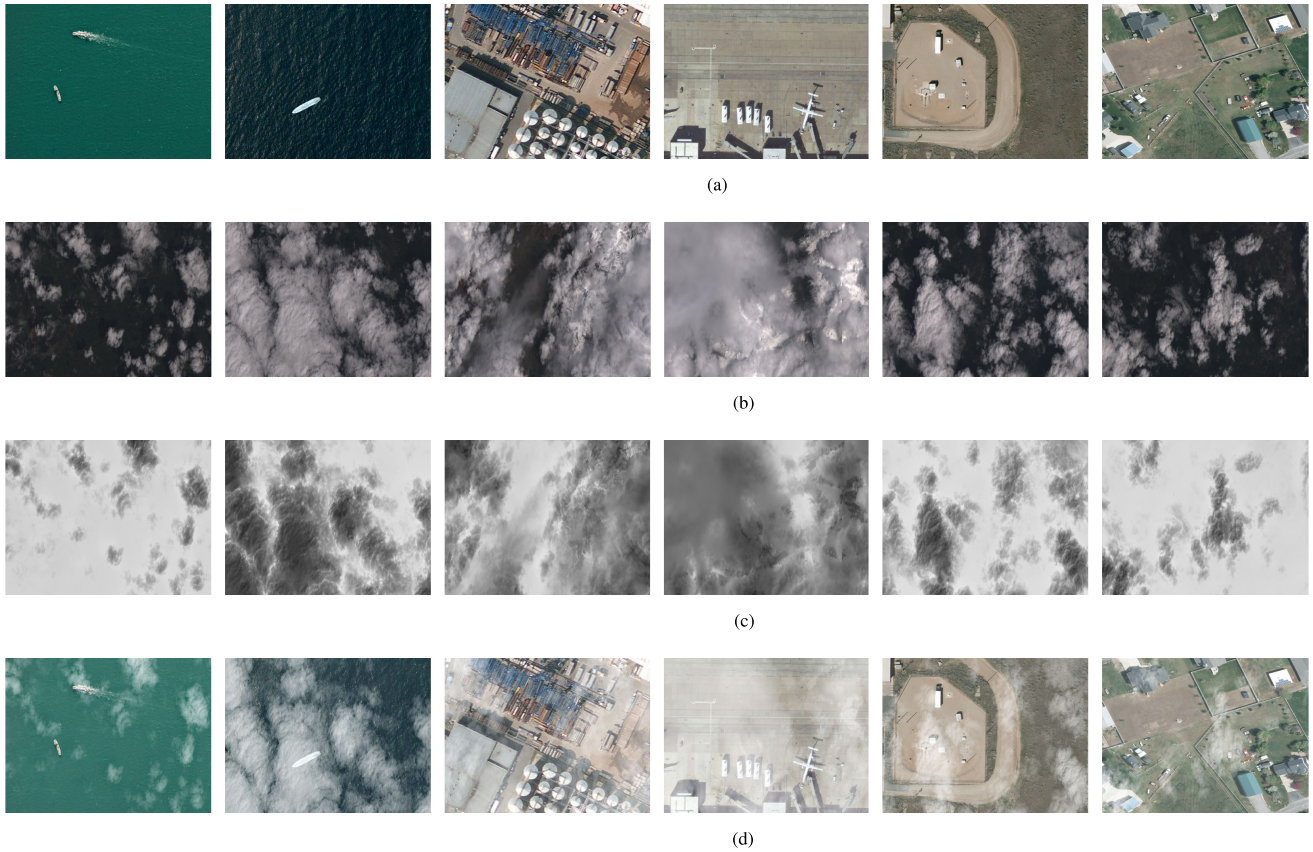


Fig. 7. Experimental results of nonuniform haze adding. (a) Example of haze-free RS images. (b) Example of real nonuniform hazy RS images. (c) Example of transmission maps extracted from nonuniform hazy RS images. (d) Example of synthesized nonuniform hazy RS images.

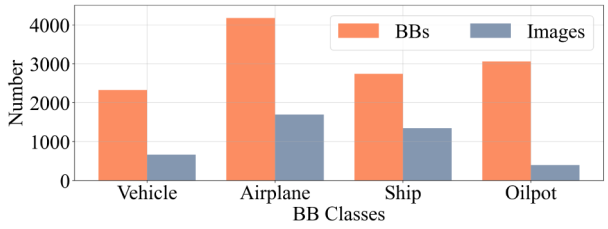


Fig. 8. Number of each type of BB and the number of images containing each type of BB.

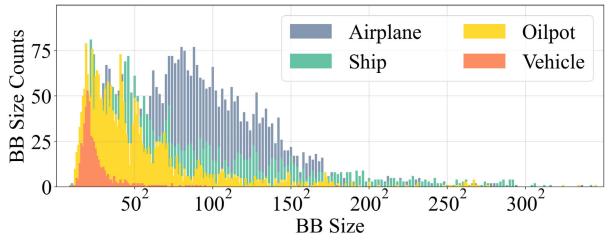


Fig. 9. Distribution of BB size of HazyRS.

this approach, the dataset also includes several small target images, as shown in Fig. 9. The four detected objects are concentrated within a distribution trend of 32×32 pixels.

IV. Experiments

A. Experimental Dataset Description

1) HazyRS Dataset: The HazyRS dataset is a hazy RS image object detection dataset synthesized from the LEVIR and VEDAI datasets. This dataset comprises 20 000 labeled RS images, which include 200 types of nonuniform haze categorized into 100 levels of haze intensity. The object detection categories are divided into four types: vehicle, airplane, ship, and oilpot. We explicitly define small targets based on absolute pixel dimensions, categorizing objects with a size smaller than 32×32 pixels as small. Given the high resolution of our imagery, this threshold implies that small targets occupy less than 0.1% of the total image area.

2) Hazy DUT Anti-UAV Dataset: To validate the effectiveness of our algorithm on low-altitude captured images, we synthesized a drone dataset in hazy weather using the DUT anti-UAV dataset [42]. This dataset contains 10 000 labeled drone images. We added varying degrees of haze to these images and adjusted the brightness to simulate adverse weather in real scenarios. When adding haze, we randomly selected global atmospheric light $A \in [0.8, 1]$ and scattering coefficient $\beta \in [0.4, 2.0]$ based on the ASM. Using the ASM, we generated the corresponding haze images. Ultimately, we obtained 10 000 drone images that can simulate shooting under hazy weather, with 5200 images allocated to the training set, 2600 images as the validation set, and 2200 images divided into the test set.

3) A2I2-Haze Dataset: To verify the generalization capability of SHRS-YOLO in real-world scenarios, we conducted experiments on the A2I2-haze dataset [43]. Unlike the synthetic datasets mentioned above, A2I2-haze is the first dataset to provide in situ smoke measurements aligned with aerial

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.



Fig. 10. Comparison of experimental results on HazyRS. (a) Example of processing results of YOLOv11. (b) Example of processing results of YOLOv11 + IDH module. (c) Example of processing results of YOLOv11 + IDH module + HP-CNN. (d) Example of processing results of YOLOv11 + IDH Module + HP-CNN + DFF module. (e) Example of processing results of SHRS-YOLO.

imagery, capturing authentic atmospheric degradation features such as nonlinear noise and blur. This dataset consists of paired hazy and haze-free images captured by uncrewed aerial vehicles (UAVs) under controlled smoke conditions. The annotated targets are primarily civilian vehicles. Due to the aerial perspective, these vehicles typically occupy extremely small spatial dimensions within the high-resolution frames. This characteristic makes A2I2-haze an ideal benchmark for evaluating the robustness of small object detection algorithms under realistic low-visibility conditions.

B. Model Training and Evaluation Metrics

1) Experimental Details: We constructed the network model for this study using PyTorch and other frameworks. Our SHRS-YOLO model was trained using the Adam optimizer for 300 epochs, with each batch consisting of 16 samples. Additionally, an NVIDIA GeForce RTX 4090 graphics card with 24 GB of video memory was employed.

2) Evaluation Metrics: The main evaluation metrics chosen for the experiment are precision (P), recall (R), and mean average precision (mAP). The formulas are as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \times 100\% \tag{25}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \times 100\% \tag{26}$$

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{27}$$

$$\mathrm{mAP} = \frac{1}{n} \sum_{i=1}^{n} \mathrm{AP}_i \tag{28}$$

where TP is the number of correctly detected objects. FP is the number of incorrectly detected objects. FN is the number of objects that were not detected. AP ranges from 0 to 1, with higher values indicating better performance. AP_i is the average precision for the ith object class. n is the total number of object classes.

C. Ablation Study

1) Ablations on the Proposed Methods: To demonstrate the effectiveness of the various modules proposed in SHRS-YOLO, we conducted ablation experiments on these components, including the IDH module, HP-CNN, DFF module, and STD-IoU loss function. The experimental results are presented in Table I, and Figs. 10 and 11. We adopted YOLOv11-M as the baseline model. Model A incorporates the manually adjusted IDH module, achieving a significant improvement of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

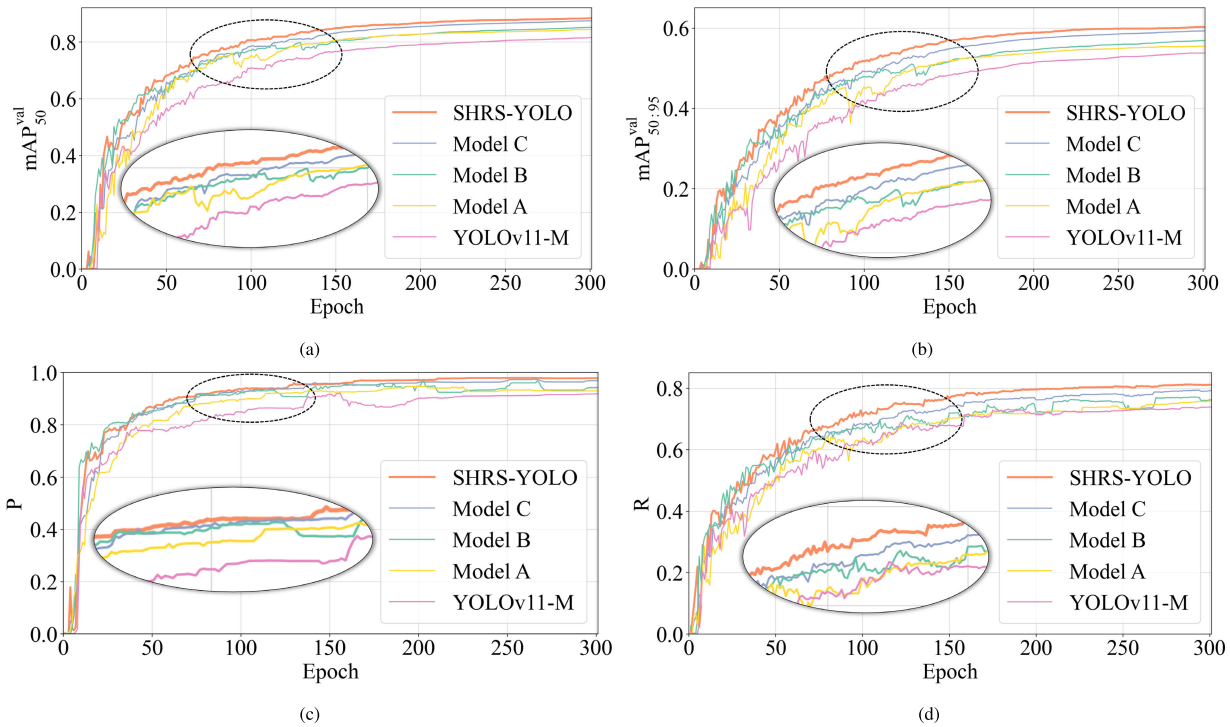


Fig. 11. Comparison of ablation experiment results for SHRS-YOLO. (a) Comparison of AP_{50}^{val}-values. (b) Comparison of $mAP_{50:95}^{val}$-values. (c) Comparison of *P*-values. (d) Comparison of *R*-values.

TABLE I
ABLATION ANALYSIS ON THE SHRS-YOLO

Model	Module				$mAP_{50}^{val}(\%)$	Param.(10^6)	FLOPs(10^9)	FPS
	IDH	HP-CNN	DFF	STD-IoU				
Base(YOLOv11-M)					81.52	20.12	68.5	176.4
Model A	✓				84.35	20.12	68.5	176.2
Model B	✓	✓			85.14	20.13	68.7	171.5
Model C	✓	✓	✓		87.48	18.32	61.4	186.7
SHRS-YOLO	✓	✓	✓	✓	**88.37**	**18.32**	**61.4**	**186.7**
Improvement					+8.40%	-8.95%	-10.36%	+5.84%

3.47% in mAP_{50}^{val} compared to YOLOv11. Model B builds upon Model A by integrating HP-CNN to learn the hyperparameters of the IDH module. The hyperparameters generated by HP-CNN enable the IDH module to perform adequate dehazing according to the haze shape and concentration of the input hazy images. Although the addition of HP-CNN increases model parameters, it significantly improves detection accuracy. Model C improves upon Model B by replacing the C2PSA in YOLOv11 with the DFF module. This replacement reduces network parameters and FLOPs by 8.99% and 10.63%, respectively, compared to Model B, while improving mAP_{50}^{val} by 2.75%. This optimization is attributed to the DFF module's efficient utilization of transmission guidance map information for comprehensive feature map fusion. Compared to Model C, SHRS-YOLO replaces the loss function with STD-IoU, improving mAP_{50}^{val} by 1.02%. This improvement is attributed to the proposed STD-IoU loss function's enhanced focus on small targets in RS images.

2) Analysis of the Input Resolution for the HP-CNN Module: To determine the optimal input size for the HP-CNN, we conducted an ablation study with input resolutions of 256 × 256, 512 × 512, and 1024 × 1024. The experimental results are summarized in Table II. As illustrated in the table, increasing the input resolution of HP-CNN does not yield a significant improvement in detection accuracy. The AP remains stable around 88.37%, and the performance across different scales shows negligible fluctuation. This phenomenon can be attributed to the fact that atmospheric light and transmission maps are global, low-frequency statistical information. A low-resolution image (256 × 256) is sufficient for the

TABLE II
ABLATION ANALYSIS OF THE INPUT RESOLUTION FOR THE HP-CNN MODULE

Resolution	AP(%)	AP_S(%)	AP_M(%)	AP_L(%)	FPS
256 × 256	88.37	81.21	89.67	94.28	186.7
512 × 512	88.37	81.21	89.67	94.28	170.1
1024 × 1024	88.38	81.22	89.68	94.28	160.5

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

TABLE III
COMPARISON OF SOTA METHODS ON THE HAZYRS DATASET

Model	Type	$\text{mAP}_{50}^{val}(\%)$	$\text{mAP}_{50:95}^{val}(\%)$	$\text{mAP}_{50}^{test}(\%)$	$\text{mAP}_{50:95}^{test}(\%)$	$\text{AP}_{\text{S}}(\%)$	$\text{AP}_{\text{M}}(\%)$	$\text{AP}_{\text{L}}(\%)$
DCP-YOLOv11 [22]	Dehaze & Detect	81.53	52.63	79.34	50.64	72.43	82.16	88.54
Grid-YOLOv11 [44]	Dehaze & Detect	81.62	53.68	79.57	51.92	72.68	82.35	88.79
MSBDN-YOLOv11 [45]	Dehaze & Detect	82.54	54.32	80.39	52.62	73.81	83.42	89.15
AOD-YOLOv11 [6]	Dehaze & Detect	82.67	54.94	80.98	52.51	74.02	83.67	89.48
AOD-FBVF-YOLO [46]	Dehaze & Detect	84.38	56.94	82.69	54.66	76.52	85.84	91.23
AOD-DMEFF-Net [47]	Dehaze & Detect	84.06	56.67	81.31	54.07	76.13	85.45	90.89
D-YOLO [48]	Union	85.31	57.37	83.67	55.91	77.46	86.92	92.15
IA-YOLO [9]	Union	84.62	57.23	83.06	55.67	76.84	86.43	91.82
MS-DAYOLO [10]	Union	85.24	57.04	83.37	55.35	77.36	86.87	92.06
DSNet [11]	Union	85.28	56.31	82.64	54.84	77.42	86.94	92.12
Cascade-Net [49]	Union	84.38	56.36	82.59	53.94	76.25	85.98	91.56
BAD-Net [50]	Union	85.25	57.54	83.04	55.85	77.28	86.82	91.93
MFAE-YOLO [51]	Union	85.54	57.96	83.85	56.07	77.85	87.24	92.36
SHRS-YOLO	**Union**	**88.37**	**60.28**	**86.32**	**58.64**	**81.21**	**89.67**	**94.28**

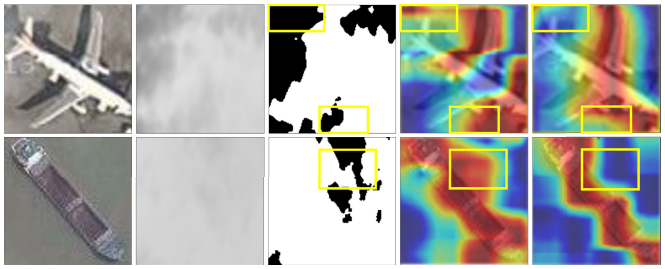


Fig. 12. Visualization of the effectiveness of the transmission-guided feature fusion. Columns from (left to right) represent. (a) Input hazy images. (b) Corresponding transmission maps. (c) Binarized transmission maps indicating haze distribution. (d) Feature heatmaps using direct concatenation. (e) Feature heatmaps using our proposed transmission-guided fusion. The yellow boxes highlight regions where the guided fusion successfully suppresses background noise and enhances target structural details compared to the unguided baseline.

HP-CNN to extract these global features and regress accurate filter parameters. High-resolution inputs merely introduce redundant spatial details that are unnecessary for hyperparameter estimation. In terms of inference speed, however, the resolution has a noticeable impact. As the resolution increases from 256 × 256 to 1024 × 1024, the additional computational overhead in the HP-CNN module causes the FPS to drop from 186.7 to 160.5, a decrease of approximately 14%. To achieve the best tradeoff between detection accuracy and real-time processing efficiency, we selected 256 × 256 as the default input resolution for the HP-CNN module in our final SHRS-YOLO framework.

3) Visualization Analysis of Transmission-Guided Fusion: To intuitively demonstrate why the transmission map can effectively distinguish the importance of local and global features, we visualized the feature activation maps in Fig. 12. As discussed in Section III-D, hazy images are characterized by nonuniform degradation: dense haze areas suffer from severe information loss and high noise, necessitating global context for recovery, while light haze areas retain sufficient local textures but are susceptible to interference from global noise. As shown in column (d), simply concatenating features without guidance leads to diffuse attention. The network fails to distinguish between the target and the background noise, causing activation to spill into nontarget regions. This confirms that without guidance, the noise in dense haze areas interferes with the representation of light haze areas. In contrast, column (e) demonstrates the efficacy of our DFF module. By utilizing the binarized transmission map [column (c)] as a spatial switch, the model explicitly suppresses background noise in dense haze regions while sharpening feature responses in light haze regions. The yellow boxes highlight that the guided features align more precisely with the target boundaries and show significantly reduced background activations. This visualization confirms that the transmission map effectively acts as a reliable prior, guiding the network to adaptively weigh local and global information based on the spatial distribution of haze.

D. Comparison With SOTA Methods

We compared the proposed method with various hazy RS image object detection algorithms on the HazyRS dataset. Compared to high-altitude RS hazy images, low-altitude hazy images exhibit more uniform haze patterns. To validate the effectiveness of SHRS-YOLO on low-altitude hazy image object detection, we also conducted experiments on the Hazy DUT anti-UAV dataset. Tables III and IV present comparative experimental results for D-YOLO [48], IA-YOLO [9], MS-DAYOLO [10], DSNet [11], Cascade-Net [49], BAD-Net [50], and our proposed method. This study also selected six two-stage dehazing-detection approaches, ranging from combinations using classical dehazing techniques [6], [22], [44], [45] to recent advanced methods such as AOD-FBVF-YOLO [46] and AOD-DMEFF-Net [47]. These two-stage pipelines typically process hazy images first using specialized dehazing models and then feed the results into pretrained detection models. The best results in the tables are highlighted in bold.

SHRS-YOLO achieved the best performance on the HazyRS dataset, attaining 88.37% $\text{mAP}_{50}^{\text{val}}$ and 60.28% $\text{mAP}_{50:95}^{\text{val}}$. Regarding detection performance across different object scales, the proposed method reached 81.21%, 89.67%, and 94.28% for AP_S, AP_M, and AP_L, respectively. Compared to MFAE-YOLO, SHRS-YOLO improved by 2.83% on $\text{mAP}_{50}^{\text{val}}$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
COMPARISON OF SOTA METHODS ON THE HAZY DUT ANTI-UAV DATASET

Model	Type	$\text{mAP}_{50}^{val}(\%)$	$\text{mAP}_{50:95}^{val}(\%)$	$\text{mAP}_{50}^{test}(\%)$	$\text{mAP}_{50:95}^{test}(\%)$	$\text{AP}_{\text{S}}(\%)$	$\text{AP}_{\text{M}}(\%)$	$\text{AP}_{\text{L}}(\%)$
DCP-YOLOv11 [22]	Dehaze & Detect	86.32	59.30	84.25	56.90	76.82	88.54	94.15
Grid-YOLOv11 [44]	Dehaze & Detect	87.98	60.58	86.56	58.39	78.43	89.76	95.02
MSBDN-YOLOv11 [45]	Dehaze & Detect	87.82	61.10	85.68	58.79	78.12	89.65	94.88
AOD-YOLOv11 [6]	Dehaze & Detect	88.17	61.78	85.87	59.54	79.44	90.36	95.20
AOD-FBVF-YOLO [46]	Dehaze & Detect	90.34	64.51	88.27	62.55	82.56	92.48	96.12
AOD-DMEFF-Net [47]	Dehaze & Detect	89.67	64.35	87.68	62.17	81.34	91.83	95.95
D-YOLO [48]	Union	92.82	65.08	90.59	62.96	86.35	94.12	97.02
IA-YOLO [9]	Union	92.13	65.01	89.98	62.56	85.48	93.63	96.88
MS-DAYOLO [10]	Union	90.84	64.85	88.42	62.75	83.15	92.74	96.45
DSNet [11]	Union	89.78	62.49	87.70	60.15	81.65	91.94	96.10
Cascade-Net [49]	Union	88.38	62.36	86.11	60.14	79.88	90.54	95.35
BAD-Net [50]	Union	91.25	64.54	88.82	62.45	84.26	93.17	96.63
MFAE-YOLO [51]	Union	92.87	65.21	89.35	62.86	86.42	94.14	97.05
SHRS-YOLO	**Union**	**94.63**	**66.35**	**92.42**	**64.17**	**89.15**	**95.84**	**98.26**



Fig. 13. Visualization of detection results on the real-world A2I2-Haze dataset.

and 2.32% on $\text{mAP}_{50:95}^{\text{val}}$. Notably, for small target detection, our method achieved a substantial improvement of 3.36% in AP_S. Compared to two-stage dehazing detection methods, SHRS-YOLO demonstrated even more significant performance advantages, outperforming the best-performing two-stage method, AOD-FBVF-YOLO, by 3.99% in $\text{mAP}_{50}^{\text{val}}$ and 4.69% in AP_S. As an end-to-end joint optimization method, SHRS-YOLO avoids the error accumulation problem caused by the separation of dehazing and detection modules in two-stage approaches.

SHRS-YOLO also achieved optimal performance on the hazy DUT anti-UAV dataset, attaining 94.63% $\text{mAP}_{50}^{\text{val}}$ and 66.35% $\text{mAP}_{50:95}^{\text{val}}$. Detailed scale-wise analysis shows that the method obtained 89.15%, 95.84%, and 98.26% for AP_S, AP_M, and AP_L, respectively. Compared to MFAE-YOLO, SHRS-YOLO improved by 1.76% on $\text{mAP}_{50}^{\text{val}}$ and 1.14% on $\text{mAP}_{50:95}^{\text{val}}$. Furthermore, our method exhibited a 2.73% gain in AP_S, validating its robustness for small targets. Compared to two-stage dehazing detection methods, SHRS-YOLO demonstrated even greater performance advantages, surpassing the best-performing AOD-FBVF-YOLO by 4.29% in $\text{mAP}_{50}^{\text{val}}$ and 6.59% in AP_S. These results indicate that SHRS-YOLO is effective not only for high-altitude RS images but also maintains superior detection capabilities for low-altitude uniform hazy images.

E. Qualitative Evaluation on Real-World A2I2-Haze Dataset

To further validate the generalization capability of SHRS-YOLO in real-world scenarios, we conducted a qualitative evaluation on the A2I2-haze dataset, which consists of aerial images with real atmospheric degradation. As shown in Fig. 13, the dataset contains small civilian vehicles obscured by nonuniform smoke and haze. Despite these challenging conditions, SHRS-YOLO effectively suppresses haze interference and accurately localizes the small targets that are barely visible to the naked eye. The visualization results demonstrate that our proposed framework generalizes well to real-world domains without requiring extensive domain adaptation.

V. CONCLUSION

We propose SHRS-YOLO, a detection framework for small objects in hazy remote-sensing images. Atmospheric scattering and absorption in hazy conditions severely degrade image contrast and color saturation, weakening feature representations of small targets and hampering target-background discrimination. Our approach incorporates an IDH module with four coordinated filters whose parameters are predicted dynamically by HP-CNN, and a DFF module that uses transmission guidance maps to adaptively fuse multiscale regional-attention features with local CNN features. SHRS-YOLO achieves

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

superior detection accuracy whilst maintaining computational efficiency—reducing parameters by 8.95%, FLOPs by 10.36%, and increasing FPS by 5.84%—demonstrating effective lightweight design without sacrificing precision. Future research will extend the framework to multispectral remote-sensing imagery and optimize the architecture for enhanced robustness under adverse weather conditions.

REFERENCES

[1] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "YOLOv9: Learning what you want to learn using programmable gradient information," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 1–21.

[2] A. Wang et al., "YOLOv10: Real-time end-to-end object detection," 2024, *arXiv:2405.14458*.

[3] R. Khanam and M. Hussain, "YOLOv11: An overview of the key architectural enhancements," 2024, *arXiv:2410.17725*.

[4] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 33, no. 12, pp. 2341–2353, Dec. 2011.

[5] Q. Zhu, J. Mai, and L. Shao, "A fast single image haze removal algorithm using color attenuation prior," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 3522–3533, Nov. 2015.

[6] B. Li, X. Peng, Z. Wang, J. Xu, and D. Feng, "AOD-Net: All-in-one dehazing network," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4780–4788.

[7] D. Engin, A. Genc, and H. K. Ekenel, "Cycle-Dehaze: Enhanced CycleGAN for single image dehazing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 825–833.

[8] B. Xiao, Z. Zheng, X. Chen, C. Lv, Y. Zhuang, and T. Wang, "Single UHD image dehazing via interpretable pyramid network," 2022, *arXiv:2202.08589*.

[9] W. Liu, G. Ren, R. Yu, S. Guo, J. Zhu, and L. Zhang, "Image-adaptive YOLO for object detection in adverse weather conditions," in *Proc. 36nd AAAI Conf. Artif. Intell.*, 2022, pp. 1792–1800.

[10] M. Hnewa and H. Radha, "Multiscale domain adaptive YOLO for cross-domain object detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2021, pp. 3323–3327.

[11] S.-C. Huang, T.-H. Le, and D.-W. Jaw, "DSNet: Joint semantic learning for object detection in inclement weather conditions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 8, pp. 2623–2633, Aug. 2021.

[12] J. Liu, B. Huang, and J. Lv, "YOLO-PKFF: Remote sensing object detection enhanced with poly kernel inception and attentional cross-level feature fusion," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[13] X. Zheng, J. Bi, K. Li, G. Zhang, and P. Jiang, "SMN-YOLO: Lightweight YOLOv8-based model for small object detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 22, pp. 1–5, 2025.

[14] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2778–2788.

[15] M. Wang et al., "FE-YOLOv5: Feature enhancement network based on YOLOv5 for small object detection," *J. Vis. Commun. Image Represent.*, vol. 90, Feb. 2023, Art. no. 103752.

[16] L. Shen, B. Lang, and Z. Song, "CA-YOLO: Model optimization for remote sensing image object detection," *IEEE Access*, vol. 11, pp. 64769–64781, 2023.

[17] Y. Zhang, M. Ye, G. Zhu, Y. Liu, P. Guo, and J. Yan, "FFCA-YOLO for small object detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611215.

[18] R. Zhang, L. Xu, Z. Yu, Y. Shi, C. Mu, and M. Xu, "Deep-IRTarget: An automatic target detector in infrared imagery using dual-domain feature extraction and allocation," *IEEE Trans. Multimedia*, vol. 24, pp. 1735–1749, 2022.

[19] R. Zhang et al., "A benchmark and frequency compression method for infrared few-shot object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5001711.

[20] R. Zhang et al., "Differential feature awareness network within antagonistic learning for infrared-visible object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 8, pp. 6735–6748, Aug. 2024.

[21] R. Zhang, L. Wu, Y. Yang, W. Wu, Y. Chen, and M. Xu, "Multi-camera multi-player tracking with deep player identification in sports video," *Pattern Recognit.*, vol. 102, Jun. 2020, Art. no. 107260.

[22] K. He, J. Sun, and X. Tang, "Single image haze removal using dark channel prior," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 1956–1963.

[23] G. Meng, Y. Wang, J. Duan, S. Xiang, and C. Pan, "Efficient image dehazing with boundary constraint and contextual regularization," in *Proc. IEEE Int. Conf. Comput. Vis.*, Dec. 2013, pp. 617–624.

[24] W. Wang, X. Yuan, X. Wu, and Y. Liu, "Fast image dehazing method based on linear transformation," *IEEE Trans. Multimedia*, vol. 19, no. 6, pp. 1142–1155, Jun. 2017.

[25] D. Chen et al., "Gated context aggregation network for image dehazing and deraining," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2019, pp. 1375–1383.

[26] Z. Zheng et al., "Ultra-high-definition image dehazing via multi-guided bilateral learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 16180–16189.

[27] J. Luo, Z. Liu, Y. Wang, A. Tang, H. Zuo, and P. Han, "Efficient small object detection you only look once: A small object detection algorithm for aerial images," *Sensors*, vol. 24, no. 21, p. 7067, Nov. 2024.

[28] S. Zhang, X. Yang, C. Geng, and X. Li, "A reparameterization feature redundancy extract network for unmanned aerial vehicles detection," *Remote Sens.*, vol. 16, no. 22, p. 4226, Nov. 2024.

[29] B. Liu, P. Mo, S. Wang, Y. Cui, and Z. Wu, "A refined and efficient CNN algorithm for remote sensing object detection," *Sensors*, vol. 24, no. 22, p. 7166, Nov. 2024.

[30] C. Wang et al., "Gold-YOLO: Efficient object detector via Gather-and-Distribute mechanism," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2023, pp. 51094–51112.

[31] J. Wang, Y. Sun, Y. Lin, and K. Zhang, "Lightweight substation equipment defect detection algorithm for small targets," *Sensors*, vol. 24, no. 18, p. 5914, Sep. 2024.

[32] S. G. Narasimhan and S. K. Nayar, "Removing weather effects from monochrome images," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.*, Jun. 2001, pp. 186–193.

[33] Y. Huang, Z.-J. Zha, X. Fu, R. Hong, and L. Li, "Real-world person re-identification via degradation invariance learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 14072–14082.

[34] A. Bhandari, A. Kafle, P. Dhakal, P. R. Joshi, and D. B. Kshatri, "Image enhancement and object recognition for night vision surveillance," 2020, *arXiv:2006.05787*.

[35] H. Ullah et al., "Light-DehazeNet: A novel lightweight CNN architecture for single image dehazing," *IEEE Trans. Image Process.*, vol. 30, pp. 8968–8982, 2021.

[36] H. Han and K. Sohn, "Automatic illumination and color compensation using mean shift and sigma filter," *IEEE Trans. Consum. Electron.*, vol. 55, no. 3, pp. 978–986, Aug. 2009.

[37] C. Wang, Y. Xiu, and Z. Chen, "Cyclone intensity prediction via multispectral image sharpening," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[38] A. Mosleh, A. Sharma, E. Onzon, F. Mannan, N. Robidoux, and F. Heide, "Hardware-in-the-loop end-to-end optimization of camera image processing pipelines," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 7526–7535.

[39] Z. Zou and Z. Shi, "Random access memories: A new paradigm for target detection in high resolution aerial remote sensing images," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1100–1111, Mar. 2018.

[40] S. Razakarivony and F. Jurie, "Vehicle detection in aerial imagery: A small target detection benchmark," *J. Vis. Commun. Image Represent.*, vol. 34, pp. 187–203, Jan. 2016.

[41] Geospatial Data Cloud. (2022). *Landsat 8 OLI-TIRS Satellite Digital Products*. [Online]. Available: http://www.gscloud.cn/sources/accessdata/411?pid=263

[42] J. Zhao, J. Zhang, D. Li, and D. Wang, "Vision-based anti-UAV detection and tracking," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25323–25334, Dec. 2022.

[43] P. Narayanan et al., "A multi-purpose realistic haze benchmark with quantifiable haze levels and ground truth," *IEEE Trans. Image Process.*, vol. 32, pp. 3481–3492, 2023.

[44] X. Liu, Y. Ma, Z. Shi, and J. Chen, "GridDehazeNet: Attention-based multi-scale network for image dehazing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 7313–7322.

[45] H. Dong et al., "Multi-scale boosted dehazing network with dense feature fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 2154–2164.

[46] X. Bai, X. Li, J. Miao, and H. Shen, "A front-back view fusion strategy and a novel dataset for super tiny object detection in remote sensing imagery," *Knowl.-Based Syst.*, vol. 326, Sep. 2025, Art. no. 114051.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.

[47] G. Zhou, L. Yu, E. Gao, and Y. Lu, "DMEFF-Net: A dynamic multiscale enhanced feature fusion model for small object detection in remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 19, pp. 1006–1022, 2026.
[48] Z. Chu, "D-YOLO a robust framework for object detection in adverse weather conditions," 2024, *arXiv:2403.09233*.
[49] D. Liu, B. Wen, J. Jiao, X. Liu, Z. Wang, and T. S. Huang, "Connecting image denoising and high-level vision tasks via deep learning," *IEEE Trans. Image Process.*, vol. 29, pp. 3695–3706, 2020.
[50] C. Li et al., "Detection-friendly dehazing: Object detection in real-world hazy scenes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 7, pp. 8284–8295, Jul. 2023.
[51] Y. Liu, X. Cheng, N. Xu, L. Wang, X. Wang, and X. Zhong, "MFAE-YOLO: Multifeature attention-enhanced network for remote sensing images object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5631214.



Runjie Yang (Student Member, IEEE) received the B.S. degree from Shenzhen University, Shenzhen, China, in 2021. She is currently pursuing the Ph.D. degree with the Faculty of Data Science, City University of Macau, Macau, China.

Her research interests include distributed system, target recognition, and image dehazing.



Yongde Guo received the B.Eng. degree from Northeastern University, Shenyang, China, in 2011, and the M.Sc. and Ph.D. degrees in information and communication engineering from Tsinghua University, Beijing, China, in 2014 and 2020, respectively.

He is currently an Associate Professor with the Faculty of Data Science, City University of Macau, Macau, China. His research interests cover image processing, pattern recognition, and spatial information processing.



Jun Yan received the Ph.D. degree from Dublin City University, Dublin, Ireland, in 1991.

He was a Foreign Academician with Russian Academy of Engineering, Moscow, Russia, and Russian Academy of Natural Sciences, Moscow, the Dean of the Mathematics and Data Science College, Qingdao University of Science and Technology, Qingdao, China, and a Special Invited Professor with the City University of Macau, Macau, China. He is currently the Chief Designer with Zhuhai No. 1 Satellite Constellation, Zhuhai Orbita Aerospace Science and Technology Company Ltd., Zhuhai, China, which consists of 34 remote-sensing satellites. His research interests include low-cost satellite design, big data processing and application, fuzzy logic control systems, artificial intelligence algorithms, and hi-rel SoC chips.



Qiuji Zhou received the bachelor's degree in data science from Zhuhai College of Beijing Institute of Technology, Zhuhai, China, in 2016, and the master's degree in data science from City University of Macau, Macau, China, in 2020, where he is currently pursuing the Ph.D. degree with the Faculty of Data Science.

In recent years, his research interests have been medical image analysis and retinopathy screening.



Haonan Yang (Student Member, IEEE) is currently pursuing the Ph.D. degree with the Faculty of Data Science, City University of Macau, Macau, China.

His research interests include intelligent transportation systems, applied cryptology, and blockchain.



Dunwei Gong (Senior Member, IEEE) received the Ph.D. degree from China University of Mining and Technology, Xuzhou, China, in 1999.

He is currently a Professor with Qingdao University of Science and Technology, Qingdao, China. As the first or corresponding author, he has authored 75 articles in CAS Zone I journals, accumulating over 10 000 Web of Science citations. His main research interests include multiobjective intelligent optimization theory, methods, and applications.

Dr. Gong received five Prizes for Natural Science. He was selected as the 2025 Shandong Taishan Scholar Distinguished Expert, a 2022 and 2023 Clarivate Highly Cited Researcher, a 2023 lifetime member of the World's Top 2% Scientists.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 18:09:42 UTC from IEEE Xplore. Restrictions apply.